
Technical and Business Writing

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Research in Computer Science

Product-Based

- Focus: Centre on creating functional software systems, applications, or platforms that solve practical problems.
- Common outcome: Software applications, Mobile apps, AI systems, Web platforms

Research-Based

- Focus: Address research problems through systematic investigation, analysis, and experimentation.
- Common outcome: Detailed analysis, experimental results, algorithm comparison.

What Information Do You Need to Show your supervisor?

What will make my project idea compelling and achievable within academic constraints?

Imagine you're pitching your final year project idea to a supervisor. What key information must you present to secure their approval?

Proposal Requirements

A research proposal is your comprehensive project plan: documenting your idea, problem, and implementation strategy.



Project Idea

Your concept and approach



Purpose

Why this project matters?



Method

How will you build or investigate?



Expected outcome

What will you deliver?

Why Research Proposals matter?

Define the research problem

Clearly articulate the specific issue you intend to address and solve

Demonstrate Feasibility

Show that your project is practical within available time and resource constraints

Review Existing Work

Identify what has already been accomplished in your research area

Why research proposals matter?

Present a Proposed Solution

Explain your approach to solving the identified problem

Plan the Project

Provide detailed methodology and realistic timeline

What is a Research Problem?



An Issue

Something that requires attention or resolution



A Question

An inquiry that needs answering



A Difficulty

A challenge that needs overcoming



A Phenomenon

An occurrence that needs explaining



A Situation

A circumstance that needs investigating

Guideline for selecting a research problem

01

Significance of the Problem

Is the problem important? Does it affect people, systems, or knowledge in meaningful ways?

02

Can It Be Framed as a Question?

A research problem should be convertible into a clear research question.

03

Can the Problem Be Narrowed?

Research problems must have clear boundaries. Avoid overly broad topics.

04

Availability of Information

Are there data sources, studies, or technologies available to support the research?

05

Researcher's Interest

If researchers aren't interested in the topic, they will lose motivation.

06

Researcher's Background

Do you have enough technical knowledge to investigate the problem effectively?

Determining the Purpose of the research

The problem: explains what the issue is

The purpose: why the research is conducted

Why is this research needed?

What gap in knowledge or practice does it address?

Who will benefit from it?

Identify the stakeholders who will gain value from the research.

What will the research achieve?

What specific outcomes or solutions will it produce?

Connection between Problem, Purpose, and Objectives

Problem

Drivers in Karachi malls struggle to find parking spaces due to inefficient parking systems that provide inaccurate information and don't guide drivers to available spots.

Purpose

To develop an **AI-based smart parking application** that **detects available parking spaces in real time** and **guides drivers directly to them** to reduce parking search time and traffic **congestion in** mall parking areas.

Writing effective problem statements

- Most weak projects fail because the problem is unclear.
- A compelling problem statement answers three critical questions:

1

What problem exists?

Define the specific issue clearly

2

Who is affected?

Identify the impacted stakeholders

3

Why are current solutions insufficient?

Explain the gap that needs filling

Writing effective problem statements

Structure of a Good Problem Statement



Need for Solution

Justify why your approach matters



Problem Gap

Identify what's missing or inadequate



Background Context

Set the scene with relevant information

Task for attendance

1. Using the guidelines discussed in the lecture, write one structured problem statement (120–150 words). Your problem statement should include:

- Background context
- Description of the problem
- Limitations of existing solutions
- Need for further research or solution

2. Develop 3–4 research objectives that explain what your project will attempt to accomplish.

- Your objectives should:
- Use clear action verbs (analyze, design, develop, evaluate, implement, etc.)
- Be specific and measurable
- Be logically connected to the problem statement